VELMÈRE BASIC AUDIT REPORT

MakerDAO Dai Stablecoin (0x6b175474e89094c44da98b954eedeac495271d0f)

VELMERE SECURITY AUDIT REPORT: MAKERDAO DAI STABLECOIN
Network: Ethereum Mainnet (Chain ID: 1)

Report ID: undefined | Tier: BASIC | Application Surface: CANONICAL

Created At: 2026-09-09T19:29:41.777Z

VERDICT SUMMARY: **LOW RISK (18/100)**

Release Decision: PASS | Verification Status: AUTOMATED_ONLY

Confidence Score: 98/100 Evidence Coverage: 98%

Proweniencja stanu: Blok #18072000 | Bytecode SHA-256: sha256:7f45c91b1... | DETERMINISTIC_REPRODUCIBLE
Merkle Root: sha256:8f49ca7bc72f7af201eed20d373856b5ae5740035ad1336be18ab0e294934fcd
Macierz pokrycia dowodowego: Bytecode 98% | CFG 97% | Funkcje 100% | Detektory 100% | Weryfikacja formalna 96%
DAI is governed by the MakerDAO protocol featuring GSM governance delay and Emergency Shutdown Module safeguards.
Note: DAI utilizes a bespoke, non-standard permit authorization scheme (allowed: bool) that predates and differs from the standardized EIP-2612 interface specification.

AUDIT OVERVIEW & CONTRACT CONTEXT

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Summary of execution scope and target objectives

Project / Contract: MakerDAO Dai Stablecoin
Contract Address: 0x6b175474e89094c44da98b954eedeac495271d0f
Network / Chain ID: Ethereum Mainnet (ID: 1)
Token / Symbol: DAI
Proxy Pattern: Immutable Token Core (Ward Auth)
Compiler Version: solc 0.5.12

SOURCE CODE VERIFICATION & BYTECODE DECOMPILATION

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

State hash match against blockchain node and dispatcher analysis

EVM machine bytecode was disassembled into a control-flow graph (CFG). Standard ERC selectors and privileged opcodes were formally analyzed.

BASELINE SECURITY FINDINGS

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Identified vulnerabilities in baseline scan

PERMISSION & ROLE GOVERNANCE MAP (PRO) PRO

[LOCKED] **[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]**

Summary of Analytical Depth in this section:

- Admin role: privilege and access control hierarchy in bytecode
- Verification of timelock delay parameters and governance controls
- Detection of privileged mint, pause, blacklist, and drainage capabilities
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

Protected evidence and findings withheld pending tier upgrade.

LIQUIDITY, HOLDER CONCENTRATION & LOCK EVIDENCE (PRO) PRO

[LOCKED] **[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]**

Summary of Analytical Depth in this section:

- Token distribution curve and top holder concentration analysis
- DEX liquidity pool pairing and lock verification across registered vaults
- Automated honeypot simulation, transfer fee checks, and sell restriction audit
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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ATTACK SURFACE & REENTRANCY FORMAL VECTORS (PRO)

PRO

[LOCKED] **[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]**

Summary of Analytical Depth in this section:

- Cross-function and cross-contract reentrancy guard modeling
- Price oracle staleness, manipulation resilience, and market feed verification
- Economic shock modeling and flash-loan manipulation resistance simulation
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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STORAGE LAYOUT COLLISION & MULTI-COMPILER BYTECODE DIFF (ADVANCED)

ADVANCED

[LOCKED] **[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]**

Summary of Analytical Depth in this section:

- Storage slot allocation collision detection across implementation upgrades
- Upgrade proxy layout gap verification and state preservation checks
- Bytecode reproduction against official compiler metadata and build pipelines
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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FORMAL SMT INVARIANT PROOFS & REGRESSION ANALYSIS (ADVANCED)

ADVANCED

[LOCKED] **[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]**

Summary of Analytical Depth in this section:

- SMT proofs for solvency, conservation, and reentrancy impossibility invariants
- Bytecode regression analysis and logic change tracking
- Formal SMT Verification (Z3): 96% invariants verified (QF_LIA UNSAT)
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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HUMAN ANALYST REVIEW & SIGNED VERIFICATION EVIDENCE (ADVANCED)

ADVANCED

[LOCKED] **[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]**

Summary of Analytical Depth in this section:

- Analyst sign-off requires verified reviewer evidence
- Cryptographically signed attestation digest
- Manual review boundary explicit: no certified safe claims
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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